

Double Integrals

ANSWER KEY



Iterated integrals over rectangles

- 1 Evaluate $\int_0^2 \int_0^3 (x^2 + y) dy dx$.
Inner: $\int_0^3 (x^2 + y) dy = 3x^2 + \frac{9}{2}$. Outer: $\int_0^2 (3x^2 + \frac{9}{2}) dx = [x^3 + \frac{9}{2}x]_0^2 = 8 + 9 = 17$.
- 2 Evaluate $\int_1^2 \int_0^1 6xy^2 dx dy$.
Inner: $\int_0^1 6xy^2 dx = 3y^2$. Outer: $\int_1^2 3y^2 dy = [y^3]_1^2 = 8 - 1 = 7$.

Double integrals over general regions

- 3 Evaluate $\int_0^1 \int_0^x (x + y) dy dx$, integrating over the triangular region $0 \leq y \leq x \leq 1$.
Inner: $\int_0^x (x + y) dy = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2} dx = [\frac{x^3}{2}]_0^1 = \frac{1}{2}$.
- 4 Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \leq x \leq 1, 0 \leq y \leq 1$.
 $\int_0^1 \int_0^1 (4 - x - y) dy dx$: inner $= 4 - x - \frac{1}{2}$. Outer: $\int_0^1 (\frac{7}{2} - x) dx = \frac{7}{2} - \frac{1}{2} = 3$.

Reversing the order of integration

- 5 Evaluate $\int_0^1 \int_0^1 e^{x^2} dx dy$ by first reversing the order of integration (the region is $0 \leq y \leq x \leq 1$).
Reversing: $\int_0^1 \int_0^x e^{x^2} dy dx = \int_0^1 x e^{x^2} dx = [\frac{1}{2} e^{x^2}]_0^1 = \frac{e-1}{2}$.

Double integrals in polar coordinates

- 6 Evaluate $\iint_D (x^2 + y^2) dA$ where D is the disk $x^2 + y^2 \leq 4$, by converting to polar coordinates ($dA = r dr d\theta$).
 $\int_0^{2\pi} \int_0^2 r^2 \cdot r dr d\theta = \int_0^{2\pi} [\frac{r^4}{4}]_0^2 d\theta = \int_0^{2\pi} 4 d\theta = 8\pi$.
- 7 Find the area of the region enclosed by $r = 2\cos\theta$ using a double integral in polar coordinates.
 $A = \int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r dr d\theta = \int_{-\pi/2}^{\pi/2} 2\cos^2\theta d\theta = \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta = \pi$.

Average value over a region

- 8 Find the average value of $f(x, y) = xy$ over the rectangle $0 \leq x \leq 2, 0 \leq y \leq 3$ (average value $= \frac{1}{A} \iint_A f dA$, where A is the region's area).
 $\iint_D xy dA = \int_0^2 \int_0^3 xy dy dx = \int_0^2 x \cdot \frac{9}{2} dx = \frac{9}{2} \cdot 2 = 9$. Area $= 6$. Average $= \frac{9}{6} = \frac{3}{2}$.

Mass and center of a region

- 9 A lamina occupies the rectangle $0 \leq x \leq 2$, $0 \leq y \leq 1$ with density $\rho(x, y) = x + y$. Find its total mass,
 $M = \iint_R \rho \, dA$.
 $M = \int_0^2 \int_0^1 (x + y) \, dy \, dx = \int_0^2 \left(x + \frac{1}{2}\right) dx = \left[\frac{x^2}{2} + \frac{x}{2}\right]_0^2 = 2 + 1 = 3$.
- 10 For a lamina occupying $0 \leq x \leq 1$, $0 \leq y \leq 1$ with constant density $\rho = 1$, find the x -coordinate of the center of mass, $\bar{x} = \frac{1}{M} \iint_R x \, dA$ (where M is the total mass).
 $M = 1 \cdot 1 = 1$. $\iint_D x \, dA = \int_0^1 \int_0^1 x \, dy \, dx = \int_0^1 x \, dx = \frac{1}{2}$. $\bar{x} = \frac{1/2}{1} = \frac{1}{2}$ (as expected, the center of a uniform square).
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Triple integral setup (a first look)

- 11 Set up (do not evaluate) the triple integral for the volume of the solid bounded by $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$.
 $V = \int_0^1 \int_0^2 \int_0^3 dz \, dy \, dx$ (which evaluates to $1 \times 2 \times 3 = 6$, the box's volume).
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Double integrals with a variable upper limit

- 12 Evaluate $\int_0^1 \int_0^{y^2} 3x \, dx \, dy$, integrating over the region $0 \leq x \leq y^2$, $0 \leq y \leq 1$.
Inner: $\int_0^{y^2} 3x \, dx = \frac{3}{2}y^4$. Outer: $\int_0^1 \frac{3}{2}y^4 \, dy = \left[\frac{3}{10}y^5\right]_0^1 = \frac{3}{10}$.
- 13 Set up and evaluate the double integral for the volume under $f(x, y) = xy^2$ over the triangular region bounded by $y = x$, $y = 0$, and $x = 2$.
 $V = \int_0^2 \int_0^x xy^2 \, dy \, dx = \int_0^2 x \cdot \frac{y^3}{3} \, dx = \int_0^2 \frac{x^4}{3} \, dx = \left[\frac{x^5}{15}\right]_0^2 = \frac{32}{15}$.
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An applied double-integral problem

- 14 A metal plate occupies the region $0 \leq x \leq 3$, $0 \leq y \leq 2$, with temperature distribution $T(x, y) = 50 - x^2 - y^2$ (degrees Celsius). Find the average temperature over the plate.
 $\iint_D T \, dA = \int_0^3 \int_0^2 (50 - x^2 - y^2) \, dy \, dx = \int_0^3 \left(100 - 2x^2 - \frac{8}{3}\right) dx = \int_0^3 \left(\frac{292}{3} - 2x^2\right) dx = \left[\frac{292}{3}x - \frac{2x^3}{3}\right]_0^3 = 292 - 18 = 274$
Area = 6. Average = $\frac{274}{6} \approx 45.67^\circ\text{C}$.
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Volume between two surfaces

- 15 Find the volume of the solid between $z = 8 - x^2 - y^2$ and $z = 0$ over the region where $z \geq 0$ requires $x^2 + y^2 \leq 8$, using polar coordinates.
 $V = \int_0^{2\pi} \int_0^{2\sqrt{2}} (8 - r^2)r \, dr \, d\theta = \int_0^{2\pi} \left[4r^2 - \frac{r^4}{4}\right]_0^{2\sqrt{2}} d\theta = \int_0^{2\pi} (32 - 16) d\theta = \int_0^{2\pi} 16 \, d\theta = 32\pi$.

Double integrals with exponential integrands

- 16** Evaluate $\int_0^1 \int_0^1 x e^{xy} dy dx$ by integrating with respect to y first.
Inner: $\int_0^1 x e^{xy} dy = [e^{xy}]_0^1 = e^x - 1$ (for $x \neq 0$). Outer: $\int_0^1 (e^x - 1) dx = [e^x - x]_0^1 = (e - 1) - 1 = e - 2$.
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A capstone applied double-integral problem

- 17** An architect is designing a swimming pool whose floor depth varies according to $d(x, y) = 1 + 0.2x + 0.1y$ (meters) over a rectangular footprint $0 \leq x \leq 10$, $0 \leq y \leq 5$ (meters). Set up and evaluate the double integral for the total volume of water needed to fill the pool, then find the average depth by dividing by the pool's surface area.
- $V = \int_0^{10} \int_0^5 (1 + 0.2x + 0.1y) dy dx$. Inner: $\int_0^5 (1 + 0.2x + 0.1y) dy = 5 + x + 1.25 = 6.25 + x$. Outer: $\int_0^{10} (6.25 + x) dx = \left[6.25x + \frac{x^2}{2}\right]_0^{10} = 62.5 + 50 = 112.5 \text{ m}^3$. Surface area = $10 \times 5 = 50 \text{ m}^2$. Average depth = $\frac{112.5}{50} = 2.25 \text{ m}$.
- 18** For the double integral $\iint_D (2x + y) dA$ where D is the region bounded by $y = x^2$ and $y = x$, complete each step:
- a** Find the intersection points of the two curves to determine the bounds. $x^2 = x$ gives $x = 0, 1$. On $[0, 1]$, $x \geq x^2$, so $x^2 \leq y \leq x$.
- b** Set up the iterated integral with x as the outer variable. $\iint_D (2x + y) dA = \int_0^1 \int_{x^2}^x (2x + y) dy dx$.
- c** Evaluate the integral. Inner: $\left[2xy + \frac{y^2}{2}\right]_{x^2}^x = \left(2x^2 + \frac{x^2}{2}\right) - \left(2x^3 + \frac{x^4}{2}\right) = \frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}$. Outer: $\int_0^1 \left(\frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}\right) dx = \left[\frac{5x^3}{6} - \frac{1}{2}x^4 - \frac{1}{10}x^5\right]_0^1 = \frac{25 - 15 - 3}{30} = \frac{7}{30}$.